

VAMSHI GARIKAPATI

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Education

University of Virginia

Master of Computer Science

Aug 2019 - May 2020

Charlottesville, VA

University of Virginia

Bachelor of Science in Computer Science — Minor in Engineering Business

Aug 2015 - May 2019

Charlottesville, VA

Work Experience

BTI360

Software Engineer II

Apr 2023 – Oct 2024

Herndon, VA

- Developed and maintained a scalable **media ETL pipeline** that extracts and segments video and audio from user-submitted links using **yt-dlp** and **FFmpeg**, triggered by **Python-based AWS Lambdas** based on link type and orchestrated through **Step Functions**.
- Monitored and analyzed backend request flows by tracking **user-linked media artifacts and metadata** in **DynamoDB**, leveraging **CloudWatch**, **CloudTrail**, **Grafana**, and **Splunk** to identify user interaction patterns and reduce incident resolution time by **35%**.
- Expanded platform capabilities by adding a new **Step Function path** to support user-requested media link types, **increasing media format compatibility by 20%** and **addressing key consumer requests**.
- Transitioned from **Step Functions** to **Lambda-triggered ECS containers** for core processing tasks, reducing media processing time by **30%**, reducing compute costs by **25%**, and improving scalability, while retaining Step Function flows for niche edge cases.
- Improved API resiliency by **30%** through redesigning **real-time visualizations** and **frontend workflows** using **VueJS** and **Python**.

Capital One

Software Engineer

Aug 2020 – Jan 2023

McLean, VA

- Developed a containerized **Plotly Dash** web application for small business **credit evaluation**, visualizing metric trends and anomalies to support faster, more accurate **risk decisions**, boosting business analyst efficiency by **40%**.
- Enhanced backend performance by implementing **threaded Snowflake caching** and building integration testing script for validating datasets
- Modernized a legacy **Python ETL package**, refactoring for compatibility with modern data tooling and improving runtime efficiency by **25%**, while enhancing code clarity and usability for data scientists.
- Integrated **New Relic** into key **EC2 instances** consuming upstream **Kafka topics** carrying real-time **customer financial events** (e.g., card swipes, account updates). Enabled monitoring of **subscription health** and ensuring **40%+ reliability** across a mission-critical data pipeline.

Capital One

Software Engineer Intern

May 2019 – Aug 2019

McLean, VA

- Developed a **RESTful mock security system** using **Angular** and **Python** to detect unauthorized access through **geographic and password reset events**, reducing breach likelihood by **10%**.
- Built a **Flask-based web platform** with D3.js to visualize **agile team survey data**, enabling **leadership** to identify key pain points and drive targeted improvements that boosted team productivity by **20–30%**.

Projects

Deep Learning Perception Architecture for Autonomous Vehicle Speed Regression

- Achieved **75% detection accuracy** and high **bounding box precision (0.8 IoU)** by training a custom **YOLOv3 model** on the **BDD100K dataset** using **Darknet** to identify real-time driving indicators.
- Streamlined model development by automating data preparation and annotation conversion with **Python**, enabling over **3000 multi-GPU training iterations** on UVA's in-house **HPC system**.

Technical Skills

Languages: Python, JavaScript, SQL, HTML5, CSS, TypeScript

Developer Tools: AWS, Terraform, Git, Jenkins, Docker, Grafana, Splunk, Chef, New Relic, Jira

Libraries/Frameworks: Plotly Dash, PostgreSQL, VueJS, Bootstrap, Flask, D3.js, Django, Angular